MIL-T-81776C(AS)
11 July 1977
Superseding
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6 December 1974

MILITARY SPECIFICATION
TEST STATION, MODULE
OJ-217( )/AWM-23(V) and
0J-378( )/AWM-23(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope, This specification covers the performance, design, and test requirements for the Module Test Station (MTS) OJ-217( )/AWM-23(V) and OJ-378( )/AWM- 23(V) (See 6.1 and 6.9).

1.2 Classification. The MTS consists of a 2-bay station and adapters as follows (see 6,4):

a, Bay No, 1: Console, Test, Module OJ-261( )/AWM-23(V)

	Nomenclature	Reference Paragraph
(1)	Control, Data Analysis C-9183( )/AWM-23(V)	3.6.1.1
(2)	Drawer Assembly, Work Surface	3.6.1.2
(3)	Power Supply Assembly PP-6858( )/AWM-23(V)	3.6.1.3
(4)	Power Supply Assembly PP-6859( )/AWM-23(V)	3.6.1.4
(5)	Power Supply Assembly PP-6860( )/AWM-23(V)	3.6.1.5
(6)	Power Supply Assembly PP-6861( )/AWM-23(V)	3.6.1.6

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:

Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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b. Bay NO. 2: Console, Test, Module OJ-260( )/AWM-23(V)

	Nomenclature	Reference Paragraph
(1)	Programmable Oscilloscope	3.6.2.1
(2)	Programmable Digital Unit	3.6.2.2
(3)	Programmable AC and DC Radiometer-Voltmeter	3.6.2.3
(4)	Programmable Counter/Timer	3.6.2.4
(5)	Photoelectric Tape Reader/ Amplifiers	3.6.2.5
(6)	Programmable Waveform Synthesizer	5.602.6
(7)	Programmable Pulse Generator	3.6.2.7
(8)	Drawer Assembly, Storage, Tool and Tape	3.6.2.8
(9)	Distribution Box J-3111( )/ AWM-23(V)	3.6.2.9

c. Adapters

	Nomenclature	
(1)	Adapters, Module	3.6.3.1
(2)	Adapter, Self Test	3.6.3.2
(3)	Adapter, Missile Auxiliaries Test Set	3.6.3.3
(4)	Adapter, Missile Interface Test Set	3.6.3.4

1.3 Associated equipment. The MTS shall operate with the cable assembly set of MIL-C-81954(AS).

2. APPLICABLE DOCUMENTS

2.1 General. The following documents, of the Issue specified in AS-40S3, Applicable Document list for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein MIL-T-81734(AS) shall be the applicable general specification for the MTS described herein (see 6.6).

SPECIFICATIONS

Military

MIL-M-7793	Meter, Time Totalizing
MIL-T-81734(AS)	Test Set, Airborne Weapon Control System, AN/AWM-23(V)
MIL-C-81954(AS)	Cable Assembly Set, Electrical, MATS, Special Purpose

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STANDARDS

Military

MIL-STD-461 Electromagnetic Interference Characteristics Requirements for Equipment

PUBLICATIONS

Hughes Aircraft Company

3282208 Acceptance Test Procedure, Module Test Station

2.2 Availability of documents, Applicable (see (3.6).

2.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply.

a. Contract. The contract shall have precedence over any other document.
b. This specification, This specification shall have precedence over all applicable subsidiary documents (see 6.6).
c* Referenced documents, Any document referenced in this specification shall have precedence over all applicable documents referenced therein,

3. REQUIREMENTS

3.1 Materials, parts, and processes. Applicable (see 6.6),

3.2 Design and construction. Applicable.

3.2.1 Design features, Applicable except as follows.

3.2.1.1 Method of operation. Applicable.

3.2.1.2 Commercial test equipment. Applicable.

3.2.1.2.1 Other commercial components. Applicable.

3.2.1.3 Circuit grounds. Applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Applicable.

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3.2.1.8 Forced-liquid cooling. Not applicable.

3.2.1.9 Space and weight limitations. The maximum dimensions shall be 48 inches by 35 inches by 75 inches with all drawers and retractable work surfaces in their retracted positions. The total weight of the MTS shall be less than 2,000 pounds.

5.2.1.9.1 Storage space. Storage space shall be provided in the MTS for special tools and program tape reels. The number and type of special tools shall be enumerated in the assembly drawing for Test Station, Module OJ-217( )/AWM-23(V) and OJ-578( )/AWM-25(V). Storage space will be provided for the module adapters of 3.4.1.4.

3.2.1.10 Connectors. Applicable.

3.2.1.11 Secondary electrical power. The secondary electrical power forms listed in Table I shall be provided by power supplies which are a part of the MTS.

3.2.1.11.1 Secondary voltage protection. Occurrence of an out-of-tolerance secondary voltage condition that could damage either an item under test or the test station shall result in automatic shutdown of the secondary power forms. The out-of-tolerance condition leading to shutdown shall be identified by an appropriate indicator. Operator action shall be required to reinstate test station operation after the occurrence of automatic shutdown.

TABLE I

SECONDARY ELECTRICAL POWER

Item	Nominal Voltage (volts)	Maximum Load Capacity [Amperes)	Location
1	+28	20	Power Supply Assembly PP-6858( )/AWM-23(V)
2	+5	20	Power Supply Assembly PP-6858( )/AWM-23(V)
3	+25	1	Power Supply Assembly PP-6859( )/AWM-23(V)
4	-25	1	Power Supply Assembly PP-6859( )/AWFl-23(V)

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3.2.1.12 Interface. The MTS shall provide the following interfaces.

3.2.1.12.1 Clean air interface. The MTS shall interface with clean air through ducts entering the top rear of each bay.

3.2.1.12.2 Electrical power interface. The MTS shall interface with electrical power for the convenience outlets at one end of the convenience outlet strip mounted on the top of the front face of the MTS, The MT'S shall interface with all other electrical power through power lines entering the top of bay number 2.

3.2.1.12.3 MATS/MITS interface. The MTS shall interface with the MATS using the MATS adapter (3.6.3.3) and the cable set of MIL-C-81954(AS). The MTS shall interface with the MITS using the MITS adapter (3.6.3.4) and the cable set of MIL-C-81954(AS).

3.2.2 Utilities. Applicable.

3.2.2.1 Electrical power. Applicable. Electrical power required for operation of the MTS shall be as listed in Table II.

3.2.2.2 Clean air. Applicable, except that clean air shall not exceed 33 pounds per minute.

3.2.2.3 Potable water. Not applicable.

3.2.2.4 Liquid coolant. Not applicable.

3.2.2.5 Helium. Not applicable.

3.2.2.6 Nitrogen. Not applicable.

3.2.2.7 Hydraulic fluid. Not applicable.

TABLE II

MTS ELECTRICAL POWER FORMS

Item	Input Power Form	Maximum Load
1	+29 +4 Vdc	0.05 kilowatt (KW)
2	115 Vac, 400 Hertz (Hz) 3-phase, 4-wire, Type III	1.0 kilovolt-amperes (kVA)
3	115 Vac, 60 Hz, 1-phase, Type 1	1.75 kVA
4	115 Vac, 60 Hz, 3-phase, 3-wire, ungrounded, Type I	2.0 kVA

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3.2.3 Service conditions.

3.2.3.1 Electrical. Applicable.

3.2.3.2 Environmental. Applicable except as follows.

3.2.3.2.1 Organizational level. Not applicable.

3.2.3.2.2 Intermediate level. Applicable.

3.2.4 Maintainability. Applicable.

3.2.4.1 Mean time to repair. Applicable.

3.2.4.2 Test points. Applicable. Test points shall be provided to facilitate isolation to the module level, fault isolation within a module, alignments or adjustments, calibration, and signal monitoring. Location and accessibility of test points shall be dictated by considerations of ease and of maintenance and repair.

3.2.4.3 Shop replaceable assembles (SRA). Applicable.

3.2.4.4 Self test capability. The MTS shall have self-test capabilities sufficient to determine its functional readiness (confidence test). The confidence test program shall be located at the beginning of each tape reel. The duration of a complete MTS confidence test shall be less than 10 minutes. Results of the confidence test shall be displayed as a GO/NO-GO on a lighted display. In addition, the MTS shall have self test capabilities sufficient to fault isolate in less than 20 minutes to a functional subassembly without disassembly. The fault isolation portion of the self-test program shall be contained on a single reel of tape and shall use the self-test adapter.

3.2.5 Reliability. The mean time between failures shall be 150 hours minimum.

3.2.5.1 Operating life. Applicable.

3.2.5.2 Storage life. Applicable.

3.2.5.3 Operational stability. Applicable.

3.2.6 Electromagnetic interference (EMI). The limits of electromagnetic interference for the MTS shall be in accordance with MIL-T-81734(AS) except as follows:

a. MIL-STD-461 Class 1D, tests CE01, CE02, CE03, and RS02-2 and the limits therein shall apply.
b. MIL-STD-461 Class 1D, test CE04 shall apply whenever power remote control leads external to the test station are used.
c. MIL-STD-461, tests RS01-1, RS02-1, and RS04 shall not apply.

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3.2.7 Utilities protection. Applicable except as follows.

3.2.7.1 Liquid cooling protection. Not applicable.

3.2.7.2 Clean air protection. Applicable.

3.2.7.3 Electrical power protection, Applicable.

3.2.7.3.1 Emergency shutdown switch. An emergency shutdown switch shall be provided for use by the operator. When the emergency shutdown switch is activated, the load side of all primary circuit breakers shall be disconnected from their respective loads. The emergency shutdown switch shall be a red push button switch located on the test monitor and control panel. The switch shall be identified by the title "EMERG OFF", and shall be marked with alternating orange-yellow and black stripes.

3.2.8 Calibration cycle. Applicable.

3.2.9 Power switching control. The MTS shall provide the necessary switching to control utility electrical power forms.

3.2.10 Displays. The MTS shall provide displays of test status, test results, test numbers, instrument in use, and power status, or malfunction.

3.3 Performance.

3.3.1 General. The MTS shall provide the capability to test and maintain certain modules of the AN/AWG-9 WCS and AN/AWM-23(V) Test Set, as specified in the assembly drawings for Test Station, Module OJ-217( )/AWM-23(V) and OJ-378( )/AWM-23(V). The MTS shall functionally test the MATS when the MT'S and MATS are connected by the MATS adapter (para. 3.6.3.3) and the cable set of MIL-C-81954(AS). The MTS shall functionally test the MITS when the MTS and the MITS are connected by the MITS adapter (Para. 3.6.3.4) and the cable set of MIL-C-81954(AS).

3.3.2 Organizational level functional requirement. Not applicable.

3.3.3 Intermediate level functional requirements. Applicable.

3.3.4 Warmup time. Applicable.

3.4 Detailed functional requirements,

3.4.1 Test capabilities. The MTS shall provide and control the stimuli, the operating power forms, and the loads required to perform functional and diagnostic testing of the modules of 3.3.1. The test capabilities provided by the MTS shall be as specified herein.

3.4.1.1 Electrical power for module testing.

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3.4.1.1.1 Fixed-voltage power. The fixed-voltage direct current (de) power forms listed in Table III shall be provided.

a. All seven of the fixed-voltage dc power supplies shall contain short circuit protective circuitry which will enable them to tolerate a load of zero ohms without damage to themselves. Upon removal of the short circuit, the supply shall regain nominal operating voltage and tolerance.
b. All seven of the fixed-voltage dc power supplies shall have fast-reaction over-voltage protective circuitry ("crowbar") to prevent any possible damage to the module under test (MUT) should the power supply fall in an over-voltage manner. The maximum reaction time shall be 10 microseconds (u sec).

3.4.1.1.2 Programmable-voltage power. Direct current voltage, programmable in one-tenth volt steps from 0.0 to 79.9 volts as specified in the following, shall be provided.

a. Each of the four programmable - voltage supplies shall deliver a full load current of 500 milliamperes (mA) at voltages between 0.0 and 60.0 volts.
b. At output voltages between 60.1 and 79.9 volts, the output full load current requirement shall be derated linearly from 500 to 150 mA.
c. Programmed values shall be accurate to +60 millivolts (mV) regardless of line, load, and temperature variations.

TABLE III

FIXED VOLTAGE POWER

Item	Voltage (volts)	Tolerance (Millivolts)	Ripple Max (mV rms)	Maximum Load Capacity (Amperes)
1	+28	±500	60	3
2	+25	± 60	20	1
3	+12	± 60	20	2
4	+ 6	± 60	20	3
5	+ 5	± 60	20	3
6	-12	± 60	20	2
7	-25	± 60	20	1

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d. Allowable ripple shall be less than 20 mV, root-mean-square (rms),

e. The power supplies shall contain short circuit protection which will enable them to tolerate a short circuit. Upon removal of the short circuit, the supplies shall regain operating voltage and tolerance.

3.4.1.2 Stimuli for module testing. The following stimuli forms shall be provided.

3.4.1.2.1 Programmable waveform synthesizer. The programmable waveform synthesizer shall provide sine waves, square waves, triangular waves and ramps in accordance with the following:

a.	Frequency	100 microhertz to 1 megahertz (MHz)
	(1) Ranges	10
	(2) Resolution	4 digits
	(3) Accuracy	+0.01 percent +0.001 Percent of range
b.	Amplitude	0.1 to 10.OV peak-to-peak (p-p)
	(1) Ranges	2
	(2) Resolution	2 digits
	(3) Accuracy	±1.0 percent +5 mV
c.	Output load	50 ohms

3.4.1.2.2 Programmable pulse generator. The programmable pulse generator shall provide pulses in accordance with the following:

a.	Pulse width	40 nanoseconds to 9.9 milliseconds
	(1) Resolution	2 digits
	(2) Ranges	6 decades
	(3) Linearity, accuracy	+5 percent full scale
b.	Rise/fall	5 nanoseconds to 7 milliseconds
	(1) Resolution	1 digit
	(2) Ranges	6 decades
	(3) Accuracy	+5 percent full scale
c.	Output amplitude	0.0V to 9.9V
	(1) Ranges	30 values
	(2) Linearity, accuracy	+5 percent full scale

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d.	Dc offset	0.0v to 9.9V
	(1) Ranges	30 values
	(2) Linearity, accuracy	+5 percent full scale
e.	Delay	O to 9.9 milliseconds
	(1) Resolution	2 digits
	(2) Ranges	6 decades
	(3) Linearity, accuracy	+5 percent full scale
f.	Source impedance	50 Ohms
g.	Pulse repetition rate (Manual control)	3 significant figures plus range select (digits O through 5 in range)
	(1) Range	9.9 Hertz (Hz) through 9.9 MHz
	(2) Linearity and accuracy	+5 percent full scale
h.	Pulse repetition rate (External control)	External trigger is required
	(1) Range	Dc to 20 MHz
	(2) Amplitude	+2 to +10V peak

3.4.1.3 Measurements for module testing. The following measurement capabilities shall be provided.

3,4. 1,3.1 Programmable ac and dc radiometer-voltmeter (digital voltmeter). The digital voltmeter shall be In accordance with the following:

a.	Dc voltage	0 to ±1099.99 volts
	(1) Resolution	6 digits
	(2) Ranges	4
	(3) Accuracy	±0.007 percent reading ±0.002 percent full scale
b.	Ac voltages	O to ±1099.99 volts rms (V, rms)
	(1) Resolution	6 digits
	(2) Ranges	4
	(3) Accuracy	±0.4 percent reading ±0.1 percent full scale

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c.	Resistance	0 to 10,999.9 kilohms
	(1) Resolution	6 digits
	(2) Ranges	5
	(3) Accuracy	±0.1 percent reading ±0.001 percent full scale ±1 ohm
d.	+Dc to +dc voltage ratios	0 to 109.999:1
	(1) Resolution	6 digits
	(2) Ranges	4
	(3) Accuracy	±0.004 percent reading ±0.001 percent (10V/REF) full scale
e.	Ac to ac voltage ratio	0 to 1099.99:1
	(1) Resolution	6 digits
	(2) Ranges	4
	(3) Accuracy	±0.1 percent reading ±0.1 percent (REF VOLT RANGE/REF) full scale
f,	Ac to +dc voltage ratio	0 to 109.999:1
	(1) Resolution	6 digits
	(2) Ranges	4
	(3) Accuracy	±0.4 percent reading ±0.1 percent (10V/REF) full scale
g.	+Dc to ac voltage ratio	0 to 1099.99:1
	(1) Resolution	6 digits
	(2) Ranges	4
	(3) Accuracy	±0.4 percent reading ±0.1 percent full scale
h.	Measurement time	Normal repetitive measurements of dc voltages, with or without polarity changes, shall be made at the rate of 100 milliseconds maximum per reading. When the filter has been called for, and polarity changes and range changes are encountered, the rate shall be less than 3 seconds maximum per reading.

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3.4.1.3.2 Programmable oscilloscope. The programmable oscilloscope shall be in accordance with the following:

a.	Vertical sensitivity	2 to 200 mV/centimeter (cm)
	(1) Ranges	7
	(2) Accuracy	+3 percent full scale (normal function) +4 percent full scale (smooth function)
b.	+Dc offset voltage	0 to ±0.995 volts
	(1) Resolution	3 digits
	(2) Ranges	10
	(3) Accuracy	+2 percent reading or 5 mV, whichever is greater
c.	Horizontal sweep	500 millisecond/cm to 100 picosecond/cm
	(1) Ranges	30
	(2) Accuracy	+3 percent full scale
d.	Input impedance	50 Ohms

3.4. 1.3.3 Programmable counter/timer. The programmable counter/timer shall be in accordance with the followlng:

a,	Time interval measurement	1 microsecond to 10^8 seconds
	(1) Resolution	0.1 microsecond to 10 seconds
	(2) Ranges	9
	(3) Accuracy (see 6.3.15)	1 count + time base accuracy ± trigger error of Channel A and Channel B
b.	Period measurement	1 microsecond to 10^8 seconds
	(1) Resolution	0.1 microsecond to 10 seconds
	(2) Ranges	9
	(3) Accuracy (see 6.3.15)	1 count + time base accuracy + trigger error
c.	Period average measurement	0.1 microsecond to 1 second
	(1) Ranges	8
	(2) Accuracy (see 6.3.15)	1 count + time base accuracy ± trigger error of + P_A periods averaged

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d.	Frequency measurement	0 to 50 MHz
	(1) Resolution	0.01 Hz to 1 MHz (as a function of range and gate interval)
	(2) Ranges	9
	(3) Accuracy (see 6.3.15)	1 count ± time base accuracy
e.	Frequency ratio (F_A/F_B)	
	(1) Ratio range	10^{-7} to 10^{7}
	(2) Frequency range	0 to 50 MHz Channel A 0 to 10 MHz Channel B
	(3) Accuracy (see 6.3.15)	± 1 count of $F_A \pm \frac{\text{trigger error of } F_B}{\text{multiplier}}$

3.4.1.4 Module adapters. Adapters shall be included with the MTS which shall provide electrical and mechanical interface between the MTS and the modules under test. Multiple-use adapters shall be used for similar modules, however, those adapters shall provide safeguards to preclude use with modules not compatible.

3.4.1.5 Programming. Programming tapes shall be used to select the proper stimulus, load, and test equipment for performing the automatic functional tests of the modules of 3.3.1. Programmable test equipment of 3.4 shall be capable of being automatically set to the proper ranges and functions by the programming tapes. A confidence and self-test program tape shall be included.

3.4.1.5.1 Photoelectric tape reader. The photoelectric tape reader shall be capable of optically reading one inch wide 8 level perforated tape and shall be in accordance with the following:

a.	Reading speed	0 to 200 characters per second in asychronous mode; 100 to 500 characters per second in free running mode; 1000 characters per second in wind/search mode.
b.	Stop distance	At 200 characters per second 0.046 inch maximum; at 500 characters per second 0.080 inch maximum; at 1000 characters per second 0.50 inch maximum.
c.	Tape storage	The reader shall accept eight inch National Association of Radio and Television Broadcasters (NARTB) reels.
d.	Read head	Silicon solar cell sensors for both code and sprocket.

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3.5 Workmanship. Applicable.

3.6 Elements. Unless otherwise specified herein, elements of the MTS will -be installed or stored in rack-and-panel type cabinets.

3.6.1 Console, Test, Module OJ-261( )/AWM-23(V). The equipment of this test console shall be in accordance with the following.

3.6.1.1 Control, Data Analysis C-9183( )/AWM-23(V). The data analysis control shall house-the MTS control logic, memory, relay matrices, and signal buffers. The control shall provide controls for module test sequencing, programming tape control, and power control. Provision shall be made on the front panel for the necessary connections and buffering to permit use of -the MTS measurement equipment for manual fault isolation, Visual displays of test status, test result, test number, instrument in use, and power status or malfunction shall also be provided.

3.6.1.2 Drawer assembly, work surface, The surface drawer assembly shall be located beneath the data analysis control. The drawer assembly shall be fully retractable, and shall provide a work surface with dimensions of 16 by 17 inches.

3.6.1.3 Power Supply Assembly PP-6858( )/AWM-23(V). The power supply assembly shall provide the following functions and features.

a. Primary power circuit breakers.
b. Primary power control circuits.
c. A +5 Vdc high current power supply which provides power to the MTS logic control circuitry, This power supply shall be provided with both short circuit and over voltage protection.
d. A +28 Vdc high current power supply which provides power to MTS relays and indicators. This power supply shall be fused to prevent overload or short circuit damage.

3.6.1.4 Power Supp ly Assembly PP-6859( )/AWM-23(V), The power supply assembly shall provide the functions and features listed below. (see 3.4.1.1.1 for characteristics of those power forms. All of the power supplies in this drawer shall contain short circuit and over voltage protection circuitry.

a. A +28 Vdc power supply which provides power to the MUT,
b. Two 25 Vdc power supplies wired to provide +25 and -25 Vdc . power to the MTS

3.6.1.5 Power Supply Assembly PP-6860( )/AWM-23(V). The power supply assembly shall provide the functions and features listed below, (see 3.4.1.3.1 fox characteristics of these power forms). Ml of the power supplies in this drawer shall contain short circuit and overvoltage protection circuitry.

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a. Two 25 Vdc power supplies wired to provide +25 and -25 Vdc power for the MUT.

b, Two 12 Vdc power supplies wired to provide +12 and -12 Vdc power to the MUT.

c. A 6 Vdc power supply to provide power to the MUT.

d. A 5 Vdc power supply to provide power to the MUT.

3.6,1.6 Power Supply Assembly PP-6861( )/AWM-23(V). The power supply assembly shall contain four 0 to 79.9 Vdc programmable power supplies (see 3.4.1.1.2) which are used to provide power to the MUT. All of these supplies shall contain short circuit protection circuitry.

3.6.2 Console, Test, Module OJ-260( )/AW14-23(V). The equipment of this test console shall be in accordance with the following.

3.6.2.1 Programmable oscilloscope. A programmable oscilloscope, Tektronix R568206J with 3S6, S1 and 3T6 plug-in units, or equivalent, shall be supplied and shall include the performance characteristics specified in 3.4,1.3.2. The oscilloscope shall be an instrument that provides a visual display of analog and digital signals. When used with a companion programmable digital unit, measurements of certain parameters of these signals shall be possible and shall be presented in digital readout form. Except for the cathode ray tube, the circuitry shall be completely solid state. The vertical and horizontal deflection systems shall be plug-in units which mount within the oscilloscope cabinet. Control of the instrument shall be possible locally through use of the front panel controls or remotely through the use of programming connectors on the rear panel. Additional rear panel connectors shall provide all signal information required by the programmable digital unit.

3.6.2.2 Programmable digital unit. A programmable digital unit, Tektronix R230206J, or equivalent, shall be supplied and shall supplement the oscilloscope of 3.6,2.1 for module testing. The programmable digital unit shall be used with the previously described oscilloscope to provide digital readout measurements of time intervals and voltage. The instrument shall be capable of local operation by front panel controls or remote operation through the use of programming connectors on the rear panel. The front panel shall contain a nixie-tube readout that displays sign, units of measure, and four digits. GO/NO-GO limit measurements shall be possible and shall be indicated by pass-fail lights on the front panel. Binary coded decimal (BCD) information of the measurements shall be provided through rear panel connectors.

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3.6.2.3 Digital voltmeter. A digital voltmeter, Dana 5743-S-4152, or equivalent, shall be supplied and shall include the performance characteristics specified in 3.4.1.3.1. The digital voltmeter shall be an instrument capable of measuring dc volts, ac volts, ohms, de-de ratio, ac-ac ratio, ac-dc ratio, and dc-ac ratio. A display of the measurement shall be presented on a five digit nixie-tube readout. A sixth digit which reads only a 0 or 1 shall extend the range of the instrument 10 percent to preserve accuracy at the range cross over points, The instrument shall be capable of being remotely programmed as well as being operated by front panel controls. Output BCD readings of the measurement values shall be available to drive output devices.

3.6.2.4 programmable counter/timer. A programmable counter/timer, Monsanto No. 110B-021B, or equivalent, shall be supplied and shall include the performance characteristics specified in 3.4.1.3.3. The counter/timer shall be capable of the following frequency and time measurements: frequency, period, average period, frequent]' ratio, totalize, and time interval. The instrument shall have a seven-digit nlxle-tube display which shall show decimal point, measurement units, and the measurement value. An overflow indicator shall indicate when the count exceeds the readout capability. The instrument shall be capable of being remotely programmed as well as being operated by front panel controls, and Will have a BCD output indication of the measurement reading,

3.6.2.5 Photoelectric tape reader. A photoelectric tape reader, Tally 600876-3 with amplifiers 10758-3, 10761-2, and 10747-2, or equivalent, shall be supplied and shall include the performance characteristics specified in 3,4.1.5.1. The tape reader shall be a hi-directional high speed unit which accepts eight inch NARTB tape reels. The read head shall contain silicon solar cell sensors for both code and sprocket output. Tape handling shall be by proportional reel servo control in conjunction with printed circuit motor single capstan drive, The tape is held against the capstan which shall be mounted directly on the motor shaft. Tape movement shall be started and stopped by current applied to the motor, Control of the instrument shall be by remote programming. Voltage-level indication of tape information shall be provided at an output connector.

3.6.2,6 Programmable waveform synthesizer. A programmable waveform synthesizer, Wavetek 157-S-131, or equivalent, shall be supplied and shall include the performance characteristics specified in 3.4.1.2.1. The waveform synthesizer shall be a highly stable source of precision sine, square, ramp, or triangular waveforms selectable over a wide frequency range. Separate control of frequency, function, and amplitude shall be provided. The instrument shall have the capability of being remotely programmed as well as being operated by front panel controls, In addition, the frequency of operatiom shall be capable of being synchronized by an external signal or frequency modulated about a pre-set center frequency if desired.

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3.6.2.7 Programmable pulse generator. A programmable pulse generator Chronetics 1012-1, or equivalent, shall be supplied and shall include the performance characteristics specified in 3.4.1.2.2. The pulse generator shall be a highly versatile source of voltage pulses. Separate control of repetition rate, pulse width, pulse amplitude, rise time, fall time, and dc offset shall be provided. The instrument shall have the capability of being remotely programmed as well as being operated by front panel controls. In addition, the instrument shall be capable of providing a burst of pulses gated by an external signal.

5.6.2.8 Drawer assembly, storage, tool and tape. The tool and tape storage drawer assembly shall provide storage for special tools and program tape reels as specified in 3.2.l.9.1. Dimensions shall be 16 inches by 24 inches by 10 inches.

3.6.2.9 Distribution Box J-3111( )/AWM-23(V). The distribution box shall provide connectors for the correct voltage and phase distribution of the three phase, 60 Hertz primary input power to the test equipment utilized in the test station.

3.6.3 Adapters. The MTS, OJ-217( )/AWM-23(V) shall include 80 adapters and the MTS, 03-378( )/AWM-23(\T) shall include 70 adapters. (see 6.9).

3.6.3.1 Module. The module adapters shall conform to the following requirements :

a. Size - 1.75 x 6.0 x 8.12 or 1.75 x 6.0 x 6.12 inches
b. Weight -Not to exceed three pounds each.

3.6.3.2 Self test. The self test adapter shall conform to the following requirements :

a. Size - 1.75 x 6.0 x 8.12 inches
b. Weight - Not to exceed three pounds.

3.6.3.3 Missile Auxiliaries Test Set. The Missile Auxiliaries Test Set adapter shall conform to the following requirements:

a. Size - 18 x 11 x 6 inches, maximum
b. Weight - Not to exceed 25 pounds.

3.6.3.4 Missile Interface Test Set. The Missile Interface Test Set adapter shall conform to the following requirements:

a. Size - 18 x 12 x 12 inches, maximum
b. Weight - Not to exceed 35 pounds.

3.6.4 Elapsed time meter. An elapsed time meter, conforming to MIL-M-7793, shall be provided. The meter shall be readily visible during operation of the MTS, and shall be capable of registering at least 9,999 hours.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable,

4.2.1 Classification. Applicable.

4.2.2 Individual items. Applicable. The Module Test Station shall include program tapes for test station self-test and functional and diagnostic programs for the applicable WCS and AN/AWM-23(V) modules,

4.2.3 In-process inspection. Applicable.

4.2.4 In-process data. Applicable.

4.2.5 Acceptance test plan. Applicable. Table IV shall apply.

4.2.6 Test results. Applicable.

4.2.7 Acceptance. Applicable.

4.2.8 Utilities. Applicable.

4.2.9 Configuration. Applicable.

TABLE IV

ACCPETANCE TEST PLAN FOR MODULE TEST STATION

OJ-217( )/AWM-23(V) and OJ-378( )/AWM-23(V)

1.0	TEST OBJECTIVE. To demonstrate the self-test and operational run-in tests of the module test station.
2.0	TEST Conditions
	a. AWG-9 Slave Units (none)
	b. Utilities Required [information only, ref. 4.2.8)
	(1) Power as specified in MIL-T-8I776(AS). (2) Forced air cooling as specified in MIL-T-81776(AS).
3.0	TESTS , In accordance with Acceptance Test Procedure 3282208
	a. Perform self-test.
	b. Perform operational run-in tests.

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4.3 Tests.

4.3.1 Electrical and electronic items - operational run-in test. Applicable.

4.3.2 Mechanical items. Applicable.

4.4 Environmental test conditions. Applicable.

4.5 Test location. Applicable.

4.6 Acceptance schedule. Applicable

4.7 Test procedures. Applicable.

4.8 Rejection and retest. Applicable.

5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The Module Test Station is intended for Intermediate level use to functionally test, fault isolate, and verify the serviceability of certain shop replaceable assemblies (SRAs) of the weapon replaceable assemblies (WRAs) of the Weapon Control System (WCS) AN/AWG-9. In addition, the MIS is intended to be used to functionally test the MATS TS-3479( )/AWM-23(V), or the MITS TS-3561( )/AWM-23(V), when the MTS and the MATS, or the MITS, are connected by the appropriate adapter (3.6.3.3 or 3.6.3.4) and the cable set of MIL-C-81954(AS).

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable. In addition, the following shall apply.

6.3.15 Programmable counter/timer accuracy. The time base stability and trigger error specified in 3.4.1.3.3 are defined as follows:

a. Time base stability

(1) crystal aging rate-less than $\pm$ 3 parts in 10^9 per 24 hours
(2) Temperature stability - $\pm$ 2 parts in 10^{10} per degree C from -20 to +55° C.
(3) Line voltage stability - Better than + 1 part in 10^7 for $\pm$ 10% variation in line voltage.

b. Trigger error - For any wave shape, trigger error is less than $\pm \left(\frac{0.0025 \text{ microseconds}}{\text{signal slope (volts/microsecond)}} \right)$.

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6.4 Nomenclature cross reference. The following is a crass reference between the official Navy nomenclature and the major contractor part numbers. The contractor part number suffix 15 controlled by the configuration list in the contract,

	Office Nomenclature	Contractor Part Number
a.	Test Station, Module OJ-217( )/AWM-23(V) and OJ-378( )/AWM-23(V)	493457-1xx
b.	Console, Test, Module OJ-261( )/AWM-23(V)	493458-1XX
c.	Console, Test, Module OJ-260( )/AWM-23(V)	493459-1XX
d.	Test Set, Missile Auxiliaries TS-3479( )/AWM-23(V)	493502-1XX
e.	Cable Assembly Set, Electrical, MATS, Special Purpose	493513-1XX
f.	Test set, Missile Interface TS-3561( )/AWM-23(V)	495507-1XX

6.5 Liquid cooler, Not applicable.

6.6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of MIL-T-81776(AS) are the same as MIL-T-81734(AS) the word “applicable” is used in this specification. When the term “not applicable” is used in this specification it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met, See 6.6 of MIL-T-81734(AS).

6.7 Abbreviations and acronyms. Applicable. In addition the following shall apply.

BCD - binary coded decimal
MA - Milliamperes
MHz - Megahertz
MTS - Module test station
MUT - Module under test
mV - Millivolts
NARTB - National Association of Radio and Television Broadcasters
rms - Root-mean-square

6.8 Convenience outlet power. Applicable.

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6.9 Alternate configurations. The OJ-378( )/AWM-23(V) configuration is the same as the OJ-217( )/AWM-25(V) configuration, except that the OJ-378 does not have the adapters to test SRAs of the AN/AWG-9 WCS Infrared Subsystem and the three card SEAM configuration. The OJ-217 ccnfiguration can be converted to the OJ-378 configuration by the deletion of the 10 SRA adapters listed in TABLE V below.

TABLE V

REFERENCE MODULE ADAPTERS

QTY	REF DES	DESCRIPTION	PART NUMBER
1	A27	Adapter (101 WRA)	3298620-2
1	A95	Adapter (101 WRA)	3298612-1
1	A17	Adapter (120 WRA)	3298618-1
1	A26	Adapter (120 WRA)	3228219-3
1	A35	Adapter (120 WRA)	3228228
1	A43	Adapter (120 WRA)	3298626-2
1	A52	Adapter (120 WRA)	3298637
1	A85	Adapter (120 WRA)	3298635
1	A19	Adapter (730 WRA)	3298676-2
1	A94	Adapter (730 WRA)	3228527

6.10 Changes from previous issue are not marked,

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